A one-parameter crack detector, measured on SEM micrographs and on pavement

DRAFT — not submitted. Every number is generated from measurement files by paper/make_paper.py; nothing in this document is typed by hand twice.

Abstract

We describe a crack segmentation detector with a single free parameter and no learned weights, and measure it on public crack data with pixel-exact ground truth in two modalities. On SEM micrographs of Inconel 718 it reaches mean per-image F1 0.4958 and IoU 0.3349, 5.8 times the F1 of marking every pixel. On a pavement benchmark, DeepCrack, the same detector reaches F1 0.7932 and IoU 0.6809. The difference between those two numbers is the paper: the detector is not worse on micrographs by accident, and the gap is not explained by annotation width, which we show by measuring it.

The detector's one constant, fitted originally on SEM steel, is optimal unchanged on the SEM alloy data and wants a different value on pavement — so it transfers within a modality and not across one. Two results cut against the method: on an SEM coating with 0.61% crack prevalence it fails outright at 0.0264 precision, and against classical detectors tuned identically it ranks first of seven on pavement but only third of seven on the SEM alloy, where Sauvola thresholding and a Sato ridge filter score as well or better. We report both rather than omit them.

Status of this draft. Section 7 lists what is missing before submission.

1 Why another crack detector

There are many. Most are learned, most report F1 or IoU against a benchmark, and most are compared against each other at their own authors' operating points. Our interest is narrow and practical: a materials laboratory needs crack measurements out of scanning-electron and transmission X-ray micrographs, on a machine with no GPU, from a tool a researcher can audit line by line and correct by hand. That rules out most of the field, and it makes the natural baseline not a network but a threshold.

The detector here is close to that baseline. It flattens the background at seven scales, takes pixels that are dark at any of them, applies a robust outlier cut and an absolute darkness floor, and cleans the result. It has one free constant, mad_k. There is no training step and no model file.

Reporting such a thing honestly turns out to be harder than building it.

2 The detector

Figure 1 is the whole path. Two gates do the detection: a robust median-absolute-deviation cut at mad_k standard deviations, and an absolute floor on intensity. Everything else is preparation and cleanup.

Two decisions are worth stating because both were made by measurement and both went against the intuition that produced the code in the first place.

The region classifier is switched off. An eight-feature classifier scored each candidate region, and deleting it improved every metric with no free parameters: paired median IoU +0.2296 over thirteen specimens, p = 0.0024. Three specimens that had scored recall 0.0000 at every candidate setting ever tried were never invisible to the proposal stage; the classifier was rejecting them.

Tiles cannot stand in for frames. Rules validated on 512 px crops deleted 40–43% of hand-marked crack when run on whole frames. The widest crack in the training crops was 43.7 px; on full frames crack widths reach 178 px, so a region rule fitted on crops is extrapolating on the object it most needs to keep.

3 On public data with pixel-exact ground truth

Three public sets, all scored by one instrument at 0 px tolerance with mad_k chosen leave-one-image-out. Image-level grouping is the right unit: these are independent fields, not repeated views of one specimen.

set	modality	material	n	crack prevalence
Inconel 718	SEM	nickel superalloy	26	4.53%
EBC	SEM	environmental barrier coating	8	0.61%
DeepCrack	optical	concrete and asphalt	237	4.33%

The two SEM sets are the crack classes of a public cross-material microstructure collection (Zenodo 23045328, CC BY 4.0). Their masks were checked rather than assumed: the coating masks are strictly binary, the alloy masks are anti-aliased binary — 94.7% of pixels exactly 0, 4.3% at 246 or above, 0.52% in between — so a mid-grey threshold is safe, and the scorer refuses to run on a mask that is not bimodal.

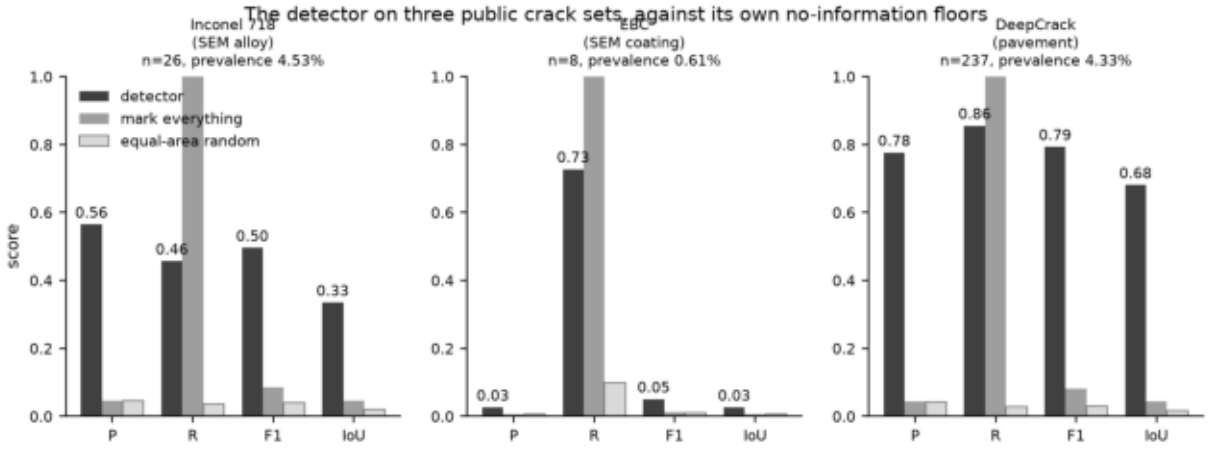


Figure 1. P1 - public scores

3.1 The detector is much worse on micrographs, and that is the result

On SEM alloy the detector reaches IoU 0.3349; on pavement, 0.6809. Both are far above their own no-information floors — 7.4 and 15.7 times the IoU of marking every pixel — so neither is a failure to detect anything. They are a factor of two apart on the same metric with the same code.

This matters beyond the benchmark. The same detector scores IoU roughly 0.33–0.42 on our own laboratory SEM corpus, which is within the range it reaches on Inconel 718 here. An independent laboratory's alloy micrographs, annotated by other people to pixel precision, put the detector in the same place our own data does. Whatever makes micrographs harder is a property of the imagery, not of our annotation.

3.2 One constant, and the limit of its transfer

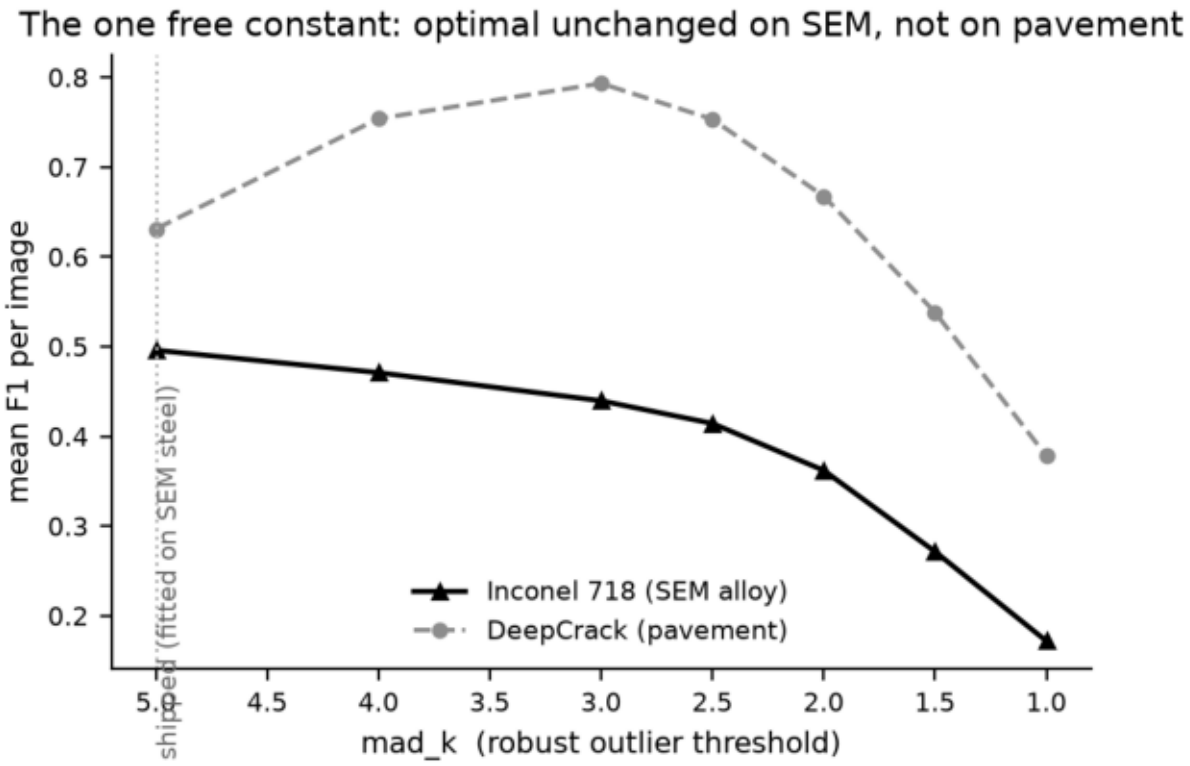


Figure 2. P2 - operating curve

The shipped constant mad_k = 5.0 was fitted on SEM steel. On the SEM alloy set it is the best of the seven values tested, and the curve falls away monotonically below it. On pavement the same detector wants a smaller value and loses 0.1621 F1 if the SEM constant is used unchanged.

So the constant transfers within a modality and not across one. That is a weaker claim than "one refit covers everything" and a more useful one for a tool: a laboratory that calibrates once on its own microscope does not recalibrate per specimen.

3.3 The failure case

On the EBC coating the detector reaches precision 0.0264 at a crack prevalence of 0.61%. Recall is high (0.7277) — the cracks are found — but the detector marks far too much besides, and at this prevalence a small absolute false-positive area destroys precision. We report it because a method paper that shows only the sets where a method works is not reporting a method.

3.4 Against classical detectors, the detector does not lead on SEM



Figure 3. P4 - baselines

Every arm gets its own best parameter by the same leave-one-image-out rule, on the same images, through the same scoring function. The result is the opposite of convenient.

On pavement this detector ranks first of seven (0.7932 against 0.7100 for the next arm). On the SEM alloy it ranks third of seven: Sauvola local thresholding reaches 0.5145 and Sato ridge 0.4985 against our 0.4958. On the SEM coating it ranks fifth of seven.

We built this detector for micrographs and it is beaten on micrographs by a local threshold from 2000 and a ridge filter from 1998. The three leading arms on Inconel 718 are separated by 0.0186 F1 — about the width of the gap we have previously called irresolvable on a corpus this size — so the honest statement is that on SEM alloy this detector and two classical filters are indistinguishable, and that the margin it enjoys on pavement does not survive the change of modality.

What survives is the comparison's premise rather than its winner: none of these arms is trained, all are tuned identically, and a reader can see what the whole classical family reaches on each set.

4 What the scores measure

This is the part we would most want another group to check.

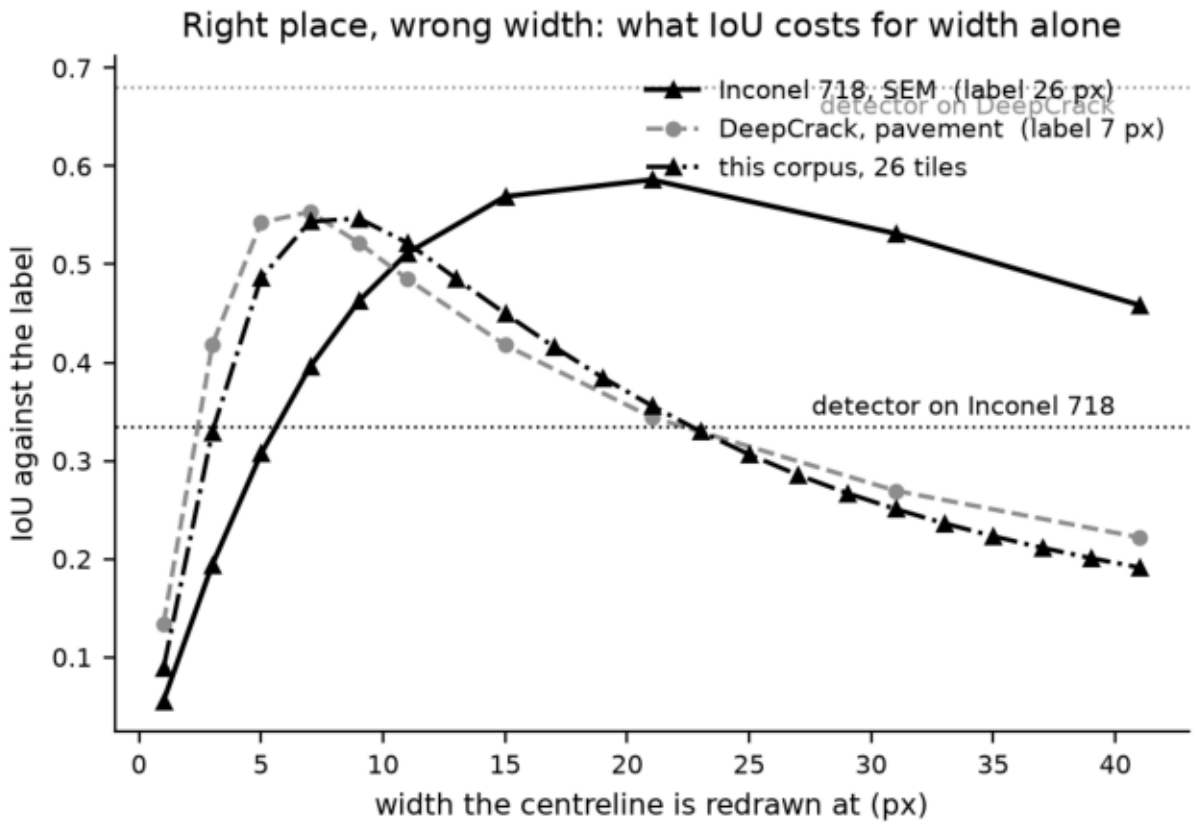


Figure 4. P3 - width sensitivity

Take the ground truth, reduce it to its centreline, and redraw it at a fixed width. The result is exactly right about where every crack is and wrong only about how wide. Scored against the label it should be nearly perfect, and it is not. Most of the available IoU is being spent on agreeing about width, and the curves peak at different widths in different corpora — so the same detector's IoU can differ between datasets without its localisation differing at all.

The SEM sets make the point sharply on their own. Inconel 718's labels are drawn at a median 25.8 px and the coating's at 6.4 px — a factor of four apart — and the best constant-width reconstruction reaches 0.5861 and 0.5797 respectively. Four times the brush width, the same ceiling to within 0.007.

This does not explain our own gap, and we initially claimed it did. The obvious reading of a detector scoring IoU 0.6809 on pavement and 0.3349 on SEM alloy is that the annotations differ in breadth. Measured, that reading fails: the constant-width curve peaks at 0.5535 on DeepCrack and 0.5467 on our own corpus tiles — within 0.01 of each other despite a large difference in brush width. Whatever makes micrographs harder for this detector, it is not how wide the labels are drawn.

The reconstruction is not an upper bound, and we presented it as one. The detector exceeds it on DeepCrack (0.6809 against 0.5535), because a real label varies in width along its length and a detector that tracks that variation beats a constant-width redraw of the same centreline. We had to be told this by the measurement.

5 Negative results

These cost the most time and are the most likely to save someone else's.

Retuning the detector for a different modality does nothing. Applying the SEM-calibrated detector to transmission X-ray tomography, we swept its two gating constants

singly and jointly, plus every morphology and size floor, under leave-one-specimen-out selection. No arm beat doing nothing on held-out data; the joint optimum lost 0.021 F1 when its selection was held out, and resolution-matching the scale ladder to the measured width ratio moved F1 by 0.0008.

That corpus cannot answer the question anyway. It holds four specimens, so a paired test has an exact two-sided floor of p = 0.125 — no result on it can reach significance at any conventional level, a win or a null. One specimen carries 57.7% of all adjudicated not-crack pixels, so the pooled score surface is essentially that specimen's.

F1 is width-gameable where the labels are broad. On the same corpus a by-construction-correct 3 px crack scores F1 0.1260, below marking every pixel (0.1499), while F1 climbs to 0.64 on marked width alone. "Beats the no-information floor on F1" is necessary and not sufficient; precision lift separates where F1 does not.

Six precision rules validated on crops were deleters on frames. See §2.

6 Measurement practices this forced

Stated as rules because each one followed a specific error of ours.

- Report the no-information floor beside every score. Marking every pixel scores F1 0.0809 on DeepCrack and 0.0857 on the SEM alloy set; at the coating's 0.61% prevalence the floor is lower still. A bare score cannot be read without it, and the floor moves by a factor of seven across these three sets.
- Choose parameters by grouped hold-out, and say what the group is. Regrouping alone moved an AUC by +0.0458 in our data, which is larger than most effects we chased.
- A difference of medians is not a paired difference. One result moved 0.0742 as medians and 0.0000 paired.
- State the resolution of the comparison. Our laboratory corpus detects about +0.14 F1 for a new method; we reported smaller differences for weeks before measuring this.
- Controls must be able to fail. Every control we built was broken on purpose to check it noticed; two did not and were rewritten.
- Check the mask encoding before scoring it. One of the two SEM crack sets stores anti-aliased masks spanning 0-255; thresholding the other set's convention onto it would have scored a large part of the ground truth as background.
- An oracle must match the arity of the thing it bounds. The constant-width reconstruction in §4 is not an upper bound, and we described it as one until the data said otherwise.

7 What this draft is missing

Listed rather than hidden.

- The SEM evidence is 26 images of one alloy and 8 of one coating, from a single released collection whose record is one sentence long and names no pixel size, no microscope and no annotation protocol. Two crack classes out of eighteen targets. That is enough to place the detector and not enough to characterise it, and a second independent SEM crack set would change this section more than anything else here.
- CrackForest was measured and removed from this draft as road imagery, like DeepCrack. Its numbers remain in docs/figures/public_metrics.json.
- The transfer claim is now within-modality only and rests on one SEM set. "The constant is optimal unchanged on SEM alloy" is one dataset agreeing with the corpus it was fitted on; it is suggestive, not established.
- No laboratory results are reported. By choice: the specimen measurements are not being published here. Sections 2 and 5 cite effects measured on that corpus as methodological evidence; a reviewer cannot check those without the data, and they should be read as weaker than the public-dataset numbers.

- The corpus series in Figure 3 carries no stroke width. The instrument that would measure it comparably needs a tile cache not present on this machine.
- No timing, and no comparison against a trained network on equal footing. A fair speed claim needs matched hardware, and an earlier attempt found this detector 1.6x slower than a single-pass U-Net at p = 0.0001.
- Novelty is not claimed. Fifty-five framings of this work were checked against prior art and all were already owned. What is offered is measurement.

Appendix A. The detector, end to end



Figure 5. Every constant read from analysis/detector_overrides.py when the plate was drawn.